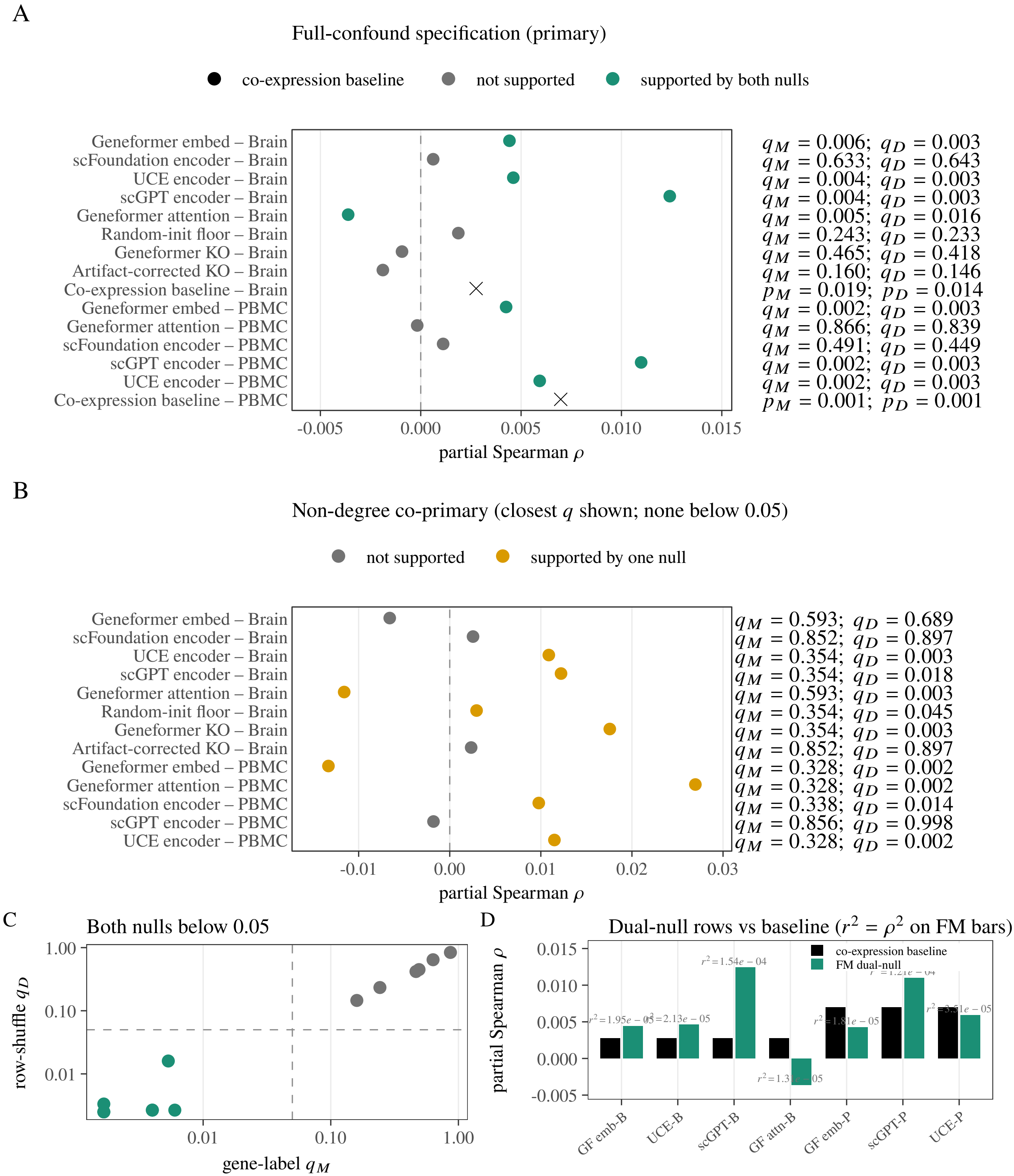

A
Full-confound specification (primary)
co-expression baseline
not supported
supported by both nulls
Geneformer embed – Brain
scFoundation encoder – Brain
UCE encoder – Brain
scGPT encoder – Brain
Geneformer attention – Brain
Random-init floor – Brain
Geneformer KO – Brain
Artifact-corrected KO – Brain
Co-expression baseline – Brain
Geneformer embed – PBMC
Geneformer attention – PBMC
scFoundation encoder – PBMC
scGPT encoder – PBMC
UCE encoder – PBMC
Co-expression baseline – PBMC
q_M = 0.006; q_D = 0.003
q_M = 0.633; q_D = 0.643
q_M = 0.004; q_D = 0.003
q_M = 0.004; q_D = 0.003
q_M = 0.005; q_D = 0.016
q_M = 0.243; q_D = 0.233
q_M = 0.465; q_D = 0.418
q_M = 0.160; q_D = 0.146
p_M = 0.019; p_D = 0.014
q_M = 0.002; q_D = 0.003
q_M = 0.866; q_D = 0.839
q_M = 0.491; q_D = 0.449
q_M = 0.002; q_D = 0.003
q_M = 0.002; q_D = 0.003
p_M = 0.001; p_D = 0.001
-0.005
0.000
0.005
0.010
0.015
partial Spearman ρ
B
Non-degree co-primary (closest q shown; none below 0.05)
not supported
supported by one null
Geneformer embed – Brain
scFoundation encoder – Brain
UCE encoder – Brain
scGPT encoder – Brain
Geneformer attention – Brain
Random-init floor – Brain
Geneformer KO – Brain
Artifact-corrected KO – Brain
Geneformer embed – PBMC
Geneformer attention – PBMC
scFoundation encoder – PBMC
scGPT encoder – PBMC
UCE encoder – PBMC
q_M = 0.593; q_D = 0.689
q_M = 0.852; q_D = 0.897
q_M = 0.354; q_D = 0.003
q_M = 0.354; q_D = 0.018
q_M = 0.593; q_D = 0.003
q_M = 0.354; q_D = 0.045
q_M = 0.354; q_D = 0.003
q_M = 0.852; q_D = 0.897
q_M = 0.328; q_D = 0.002
q_M = 0.328; q_D = 0.002
q_M = 0.338; q_D = 0.014
q_M = 0.856; q_D = 0.998
q_M = 0.328; q_D = 0.002
-0.01
0.00
0.01
0.02
0.03
partial Spearman ρ
C
Both nulls below 0.05
row-shuffle q_D
1.00
0.10
0.01
0.01
0.10
1.00
gene-label q_M
D
Dual-null rows vs baseline ($r^2 = \rho^2$ on FM bars)
partial Spearman ρ
0.015
0.010
0.005
0.000
-0.005
co-expression baseline
FM dual-null
$r^2 = 1.95e-05$
$r^2 = 2.13e-05$
$r^2 = 1.54e-04$
$r^2 = 1.81e-05$
$r^2 = 3.51e-05$
GF emb-B
UCE-B
scGPT-B
GF attn-B
GF emb-P
scGPT-P
UCE-P